

Dark-field CT Super-resolution Using Grid-anchored Gaussian Primitives

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Abstract

Gaussian primitives constitute a spatially continuous representation method modeling 3D arrays as sets of ellipsoidal kernels with learnable geometry and intensity. This work investigates their application to super-resolution (SR) in dark-field computed tomography (DFCT). To this end, we trained a convolutional neural network (CNN) to predict a 3D representation from low-resolution (LR) volumes, which we use to render high-resolution (HR) voxels. On 20 test scans with synthetic fourfold axial in-plane degradation, we find that a sparse set of 640k Gaussians is already sufficient to render 4×4 upscaled HR reconstructions at 37.14 dB peak signal-to-noise ratio (PSNR), closely matching a PixelShuffle CNN baseline at 37.34 dB. Our results demonstrate that CNN-driven parameterization of Gaussian primitives provides a viable strategy for SR in DFCT, offering a compact continuous representation rather than grid-locked voxels.

1 Introduction

Grating-based X-ray dark-field imaging provides microstructure-sensitive scattering contrast [1]. Recently, DFCT has been realized at the human scale with a dedicated gantry prototype [2]. However, due to the continuous rotation of the setup, reconstruction is highly ill-posed and image quality diminished [3]. Average-pooling can be used as a simple means to improve noise statistics in DFCT reconstructions, albeit at the cost of losing structural details. This motivates the development of advanced SR methods capable of recovering HR information after sub-sampling.

Gaussian primitives provide explicit, efficient, and physics-guided representations for images [4]. These representations have been explored for conventional attenuation-based computed tomography, providing feasible reconstruction quality even at sparse-view conditions [5]. Additionally, their formal resemblance to point spread functions offers great potential for SR in medical imaging [6]. In this work, we explore combining CNN-mediated feature extraction with the benefits of flexible Gaussian representations for SR in DFCT.

2 Methods

Data. We split an in-house DFCT dataset into 85 training, 4 validation, and 20 test volumes. Training and validation sets included a mix of ex vivo porcine lungs and various phantom materials. The held-out test set comprised a separate porcine lung placed inside of an anthropomorphic Lungman thorax phantom scanned at varying positions. DFCT volumes comprised 16 axial slices, which we min-max normalized, anti-aliased with an isotropic Gaussian kernel ($\sigma = 0.8$) and sub-sampled in-plane by a factor of 4 via average-pooling. Thereby, we synthesized $16 \times 200 \times 200$ LR volumes from the original $16 \times 800 \times 800$ HR data.

Model. We trained a CNN to infer Gaussian parameters from LR input volumes. These grid-anchored Gaussians were then queried in local neighborhoods to render a HR representation. The full *Vol2Gauss* framework is illustrated in Fig. 1.

LR signal and discrete gradients were fed to a 3D CNN mapping a $16 \times 200 \times 200$ LR volume to a $16 \times 200 \times 200 \times 12$ parameter field, equivalent to 640k Gaussians with 12 attributes each. These attributes encompassed one intensity correction, three center offsets, three scaling factors, four parameters for

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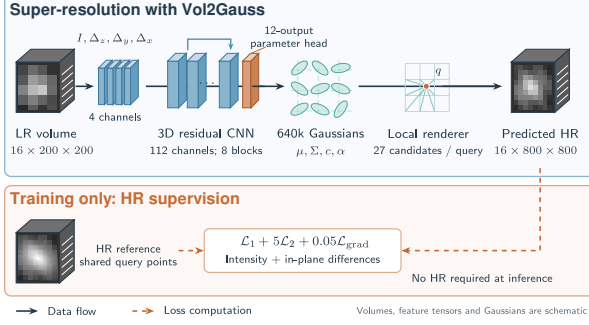


Figure 1. DFCT super-resolution framework.

a quaternion rotation and one weight. In order to upscale this representation, the primitives were queried within parent cells of size $3 \times 3 \times 3$ on the $16 \times 800 \times 800$ HR grid. For each query q , our renderer collects enclosed Gaussians and yields the predicted HR signal:

$$\hat{V}(q) = \frac{\sum_i w_i(q) c_i}{\max(\sum_i w_i(q), \varepsilon)}, \quad (1)$$

$$w_i(q) = \alpha_i e^{-\frac{1}{2}(q - \mu_i)^T \Sigma_i^{-1} (q - \mu_i)}. \quad (2)$$

where c_i, α_i, μ_i and Σ_i denote intensity, weight, center and covariance. Instead of visibility-based alpha compositing, we thereby used scalar averaging to render new voxels for the HR grid.

Evaluation. We computed PSNR, structural similarity index measure (SSIM) [7], and high-frequency error norm (HFEN) [8] to assess SR performance across the test set. Bicubic interpolation served as the weak conventional baseline, a 48-channel voxel CNN with six channel-attention residual blocks and PixelShuffle [9] predicting a residual over bicubic upsampling as the strong baseline.

3 Experiments

Representation capacity. To examine sparser representations as prediction budget, we used $16 \times 50 \times 50$, $16 \times 100 \times 100$ and $16 \times 200 \times 200$ anchor grids (40k/160k/640k Gaussians), fitted directly to each of the test volumes. Fitting was realized with 1000 steps, 32,768 queries each, and optimized per the L_1 loss function.

Rendering query density. In addition to the 1-child model (640k Gaussians), we tested a denser variant predicting 16 independent children per LR cell, amounting to a $16 \times 200 \times 200 \times 16 \times 12$ parameter field (10.24M Gaussians). For this comparison, models were trained for 80 epochs and checkpoints were selected by maximum validation PSNR.

CNN integration. Our *Vol2Gauss* model is based on a CNN mapping LR voxel space to Gaussians parameters. As alternative formulation, we fitted Gaussians to LR volumes prior to feeding them to

the parameter-predicting CNN, which we refer to as *Gauss2Gauss*. Gaussians were fitted analogously to the representation capacity experiment.

4 Results and Discussion

Gaussian primitives can represent the target HR volumes substantially better as representation capacity increases, demonstrated by the values in Tab. 1.

Table 1. Capacity of Gaussian primitives for HR fitting ($n = 20$, mean $\pm$ standard deviation).

Gaussians	PSNR $\uparrow$ (dB)	SSIM $\uparrow$	HFEN $\downarrow$
40k	34.64 ± 0.99	0.87 ± 0.02	1.01 ± 0.03
160k	36.53 ± 1.08	0.90 ± 0.02	0.80 ± 0.05
640k	38.80 ± 1.01	0.95 ± 0.01	0.61 ± 0.03

Tab. 2 lists SR performance metrics, indicating that *Vol2Gauss* outperforms the *Gauss2Gauss* formulation. The 16-child 10.24M-Gaussian model improves performance marginally over the 1-child 640k-Gaussian variant. While the former matches the voxel CNN baseline, generation-plus-render time increases from 26 to 35 seconds per volume.

As illustrated in Fig. 2, all SR methods render predictions with mitigated streak artifacts after combined binning and upscaling. Learning-based predictions are virtually indistinguishable from each other, designating the 1-child *Vol2Gauss* model as most efficient choice for flexible DFCT SR.

The gap between prediction quality and representational capacity of Gaussians suggests remaining headroom using Gaussian representation, although there might be hard limitations due to the ill-posedness of SR. For DFCT, continued optimization must balance the suppression of streak artifacts against the preservation of structural details.

Table 2. DFCT super-resolution test metrics ($n = 20$, mean $\pm$ standard deviation). 1c: 1-child; 16c: 16-child.

Method	PSNR $\uparrow$ (dB)	SSIM $\uparrow$	HFEN $\downarrow$
Bicubic	33.88 ± 0.91	0.88 ± 0.02	0.82 ± 0.02
Voxel CNN	37.34 ± 1.29	0.90 ± 0.02	0.60 ± 0.06
Vol2Gauss (1c)	37.14 ± 1.29	0.90 ± 0.02	0.62 ± 0.06
Vol2Gauss (16c)	37.34 ± 1.28	0.90 ± 0.02	0.60 ± 0.06
Gauss2Gauss	36.03 ± 1.04	0.90 ± 0.02	0.68 ± 0.05

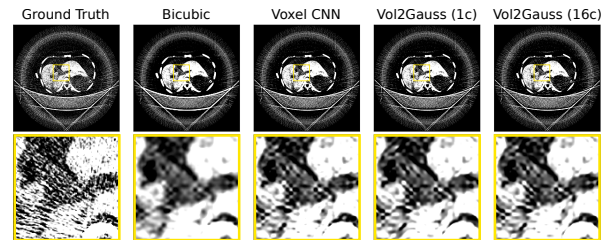


Figure 2. Exemplary DFCT super-resolution results. Top: axial slice; bottom: enlarged region of interest (120×120 pixels); 1c: 1-child; 16c: 16-child.

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